

```
### It's always a good idea when working in RStudio to start with a clean environment.

### Clear objects from the environment memory that may be leftover from previous versions of your code or
rm(list = ls())

### Load the libraries you need

library(tidyverse)
```

Warning: package 'tidyverse' was built under R version 4.5.2

Warning: package 'dplyr' was built under R version 4.5.2

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.1.4      ✓ readr      2.1.5
✓ forcats    1.0.0      ✓ stringr    1.5.2
✓ ggplot2    4.0.0      ✓ tibble     3.3.0
✓ lubridate  1.9.4      ✓ tidyr      1.3.1
✓ purrr      1.1.0

— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
! Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(dplyr)
library(car)
```

Warning: package 'car' was built under R version 4.5.2

Loading required package: carData

Warning: package 'carData' was built under R version 4.5.2

Attaching package: 'car'

The following object is masked from 'package:dplyr':

recode

The following object is masked from 'package:purrr':

some

```
library(psych)
```

Warning: package 'psych' was built under R version 4.5.2

Attaching package: 'psych'

The following object is masked from 'package:car':

logit

The following objects are masked from 'package:ggplot2':

%+%, alpha

```
#setwd("C:/Users/user/OneDrive - iagent4me.com/Desktop/Data Analytics in R")

### Load the Data
### Load the data file

brf <- read_csv("C:/Users/user/OneDrive - iagent4me.com/Desktop/Data Analytics in R/brfss2021.csv")
```

Rows: 438693 Columns: 209

— Column specification —

Delimiter: ","

chr (2): IMONTH, IDAY

dbl (196): _STATE, IYEAR, DISPCODE, PVTRES1, NUMADULT, NUMMEN, NUMWOMEN, RE...

lgl (11): COLGHOUS, TOLDCFS, HAVECF, WORKCF, TREHEPC, PRIRHEPC, HAVEHEPC...

- i Use `spec()` to retrieve the full column specification for this data.
- i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
## This study examines whether physical activity level, employment status, and educational level are assoc

#Cleaning
#1. Created a new dataframe to include only the variables needed for this analysis. Limited the original d

brf1 <- brf |>
  select("MENTHLTH", "EMPLOY1", "EDUCA", "PHYSHLTH") |>
  as.data.frame()

#NAs of each of the 4 variables in the brf1 dataframe, which aren't many
summary(brf1)
```

MENTHLTH	EMPLOY1	EDUCA	PHYSHLTH
Min. : 1.00	Min. :1.000	Min. :1.000	Min. : 1.00
1st Qu.:15.00	1st Qu.:1.000	1st Qu.:4.000	1st Qu.:25.00
Median :88.00	Median :2.000	Median :5.000	Median :88.00
Mean :59.92	Mean :3.804	Mean :5.035	Mean :63.19
3rd Qu.:88.00	3rd Qu.:7.000	3rd Qu.:6.000	3rd Qu.:88.00
Max. :99.00	Max. :9.000	Max. :9.000	Max. :99.00
NA's :2	NA's :3588	NA's :5	NA's :3

```
#Now have to address values in each of the 4 variables where a respondent answered "Don't know/Not sure or
#Need to see if removing these values from each of the variables will #significantly impact the summary st
#to do this, I created a new dataframe which converted "Don't know/Not sure or "Refused" to NAs.
#Will create a new dataframe that will impute all the NAs of my quantitative variables (MENTHLTH and PHYSH
#I will check to ensure that the distribution of the MENTHLTH and PHYSHLTH variable doesn't change dramati

#First create a new dataframe that replaces NAs for any respondent who answered Don't know/Not sure or "Re
brf2<- brf1 |>
  mutate(
```

```

MENTHLTH = case_when(
  MENTHLTH == 88 ~ 0,
  MENTHLTH %in% c(77, 99) ~ NA_real_,
  TRUE ~ as.numeric(MENTHLTH)
),
PHYSHLTH = case_when(
  PHYSHLTH == 88 ~ 0,
  PHYSHLTH %in% c(77, 99) ~ NA_real_,
  TRUE ~ as.numeric(PHYSHLTH)
),
EMPLOY1 = ifelse(EMPLOY1 == 9, NA, EMPLOY1),
EDUCA = ifelse(EDUCA == 9, NA, EDUCA)
)

```

#Now I have a second dataframe brf2, which is a copy of brf1, but will drop all the NAs. I have these 2 da

```

brf2_no_NA <- brf2 |>
  drop_na(MENTHLTH, PHYSHLTH, EMPLOY1, EDUCA)

```

```

#(brf2_no_NA)

```

#Impute NAs to the mean value of the quantitative variables, so I can compare what happens if there are no

```

brf2_imputed <- brf2 |>
  mutate(
    MENTHLTH = as.numeric(MENTHLTH),
    PHYSHLTH = as.numeric(PHYSHLTH),
    MENTHLTH = if_else(is.na(MENTHLTH), mean(MENTHLTH, na.rm = TRUE), MENTHLTH),
    PHYSHLTH = if_else(is.na(PHYSHLTH), mean(PHYSHLTH, na.rm = TRUE), PHYSHLTH)
  )

```

#The mean without the NA in MENTHLTH is approx 4.1

#the mean with imputed values in MENTHLTH is approx 4.1

```

mean(brf2_no_NA$MENTHLTH)

```

[1] 4.095012

```
mean(brf2_imputed$MENTHLTH)
```

```
[1] 4.104189
```

```
#The mean without the NA in PHYSHLTH is approx 3.77  
#the mean with imputed values in PHYSHLTH is approx 3.80  
mean(brf2_no_NA$PHYSHLTH)
```

```
[1] 3.77382
```

```
mean(brf2_imputed$PHYSHLTH)
```

```
[1] 3.795398
```

```
#The standard deviation without NAs in MENTHLTH is approx approximately 8.13  
#the mean with imputed values in MENTHLTH is approx 8.09  
sd(brf2_no_NA$MENTHLTH)
```

```
[1] 8.139498
```

```
sd(brf2_imputed$MENTHLTH)
```

```
[1] 8.091444
```

```
#The standard deviation without NAs in PHYSHLTH is approx approximately 8.32  
#the mean with imputed values in PHYSHLTH is approx 8.26  
sd(brf2_no_NA$PHYSHLTH)
```

```
[1] 8.323032
```

```
sd(brf2_imputed$PHYSHLTH)
```

```
[1] 8.262393
```

```
#I now have NAs for the categorical variables EMPLOY1 and EDUC
#Here using mean() I can see that 1.88% of the EMPLOY1 values are NA and .56% of the EDUCA values are NA.
```

```
mean(is.na(brf2$EMPLOY1)) * 100
```

```
[1] 1.880814
```

```
mean(is.na(brf2$EDUCA)) * 100
```

```
[1] 0.5648597
```

```
#This is a very low percentage, so here I will remove these NAs, since these variables will be converted to factors
brf2_imputed_no_NAs <- brf2_imputed |>
```

```
  drop_na(EMPLOY1, EDUCA)
```

```
#print(brf2_imputed_no_NAs)
```

```
#Now I will move on to additional cleaning the rest of the brf2_imputed_no_NAs dataframe because I will be converting to factors
```

```
#Convert coded categorical variables, EMPLOY1 and EDUCA to factors with new labels. I will also rename the variables
brf2_imputed_no_NAs <- brf2_imputed_no_NAs |>
```

```
  mutate(employment = factor(EMPLOY1, levels = 1:8,
                                labels = c("wage_employee",
                                             "self_employed",
                                             "unemployed_<1yr",
                                             "unemployed>1yr",
                                             "homemaker",
                                             "student",
                                             "retired",
                                             "unable_to_work"))) )
```

```
brf2_imputed_no_NAs <- brf2_imputed_no_NAs |>
```

```
  mutate(
    education = factor(
      EDUCA,
      levels = 1:6,
      labels = c(
```

```

    "no_school",
    "less_HS",
    "someHS",
    "HS_GED",
    "some_college",
    "grad_college"
  )
)
)

#print(brf2_imputed_no_NAs)

#Recode level "88" in the MENTHLTH and PHYSHLTH values as 0, to represent 0 days of reported poor mental c

brf2_imputed_no_NAs<- brf2_imputed_no_NAs |>
  mutate(mental_health_days = ifelse(MENTHLTH == 88,
                                     0,
                                     MENTHLTH),
         physical_health_days = ifelse(PHYSHLTH == 88,
                                     0,
                                     PHYSHLTH))

#print(brf2_imputed_no_NAs)
#Create a new dataframe with a shorter name to include newly cleaned variables
#this will make it easier when writing syntax in R.
#rounded to the nearest whole number

brf_clean <- brf2_imputed_no_NAs |>
  select(mental_health_days, physical_health_days, employment, education) |>
  mutate(
    mental_health_days = round(mental_health_days, 0),
    physical_health_days = round(physical_health_days, 0)
  ) |>
  as.data.frame()

#print(brf_clean)

```

The four variables chosen for this project are: MENHLTH(mental_health_days), EMPLOY1(employment), EDUCA(education), PHYSHLTH(physical_health_days)

Meaning of the variables:

mental_health_days - Indicates how many days, during the past 30 days, a survey participant considered their mental health, including stress, depression, and problems with emotions, as "not good".

employment - Indicates a survey participant's employment status.

education- Provides the highest grade or level of education completed by a survey participant.

physical_health_days - During the past month, indicates whether or not a survey participant considered their physical health "not good" due to physical illness and injury.

Variable Levels

-mental_health_days and physical_health_days 0-30: The level # corresponds with the number of days, during the past 30 days, a survey participant considered their mental health, including stress, depression, and problems with emotions, as "not good".

-employment 1. wage_employee 2. self-employed 3. unemployed_<1yr 4. unemployed>1yr 5. homemaker 6. student 7. retired 8. unable_to_work

-education 1. no_school (completed kindergarten or less,including no schooling) 2. less_HS (completed at least grades 1-8) 3. some_HS (completed at least grades 9-11) 4. HS-GED (graduated highschool or has a GED) 5. some_college (but did not graduate with a bachelor's degree college) 6. grad_college (graduated college and beyond)

Response Variable is mental_health_days (before cleaning MENTHLTH)

Before cleaning, there are 438693 responses for this variable. Here is the breakdown by level: - 36% of the survey responses indicated at least 1 poor mental health day in a month. - 62% of the survey responses indicated 0 poor mental health days in a month. - < 1% survey respondents indicated "Don't know/Not sure". - < 1% of the survey responses refused to answer. - 2 NAs(before imputation). This accounts for < 1% of the survey responses.

This variable was selected because quality of mental health is a frequently researched, reported on, and discussed topic, not solely in the medical world, but publicly. Mental health effects a person's quality of life, productivity and

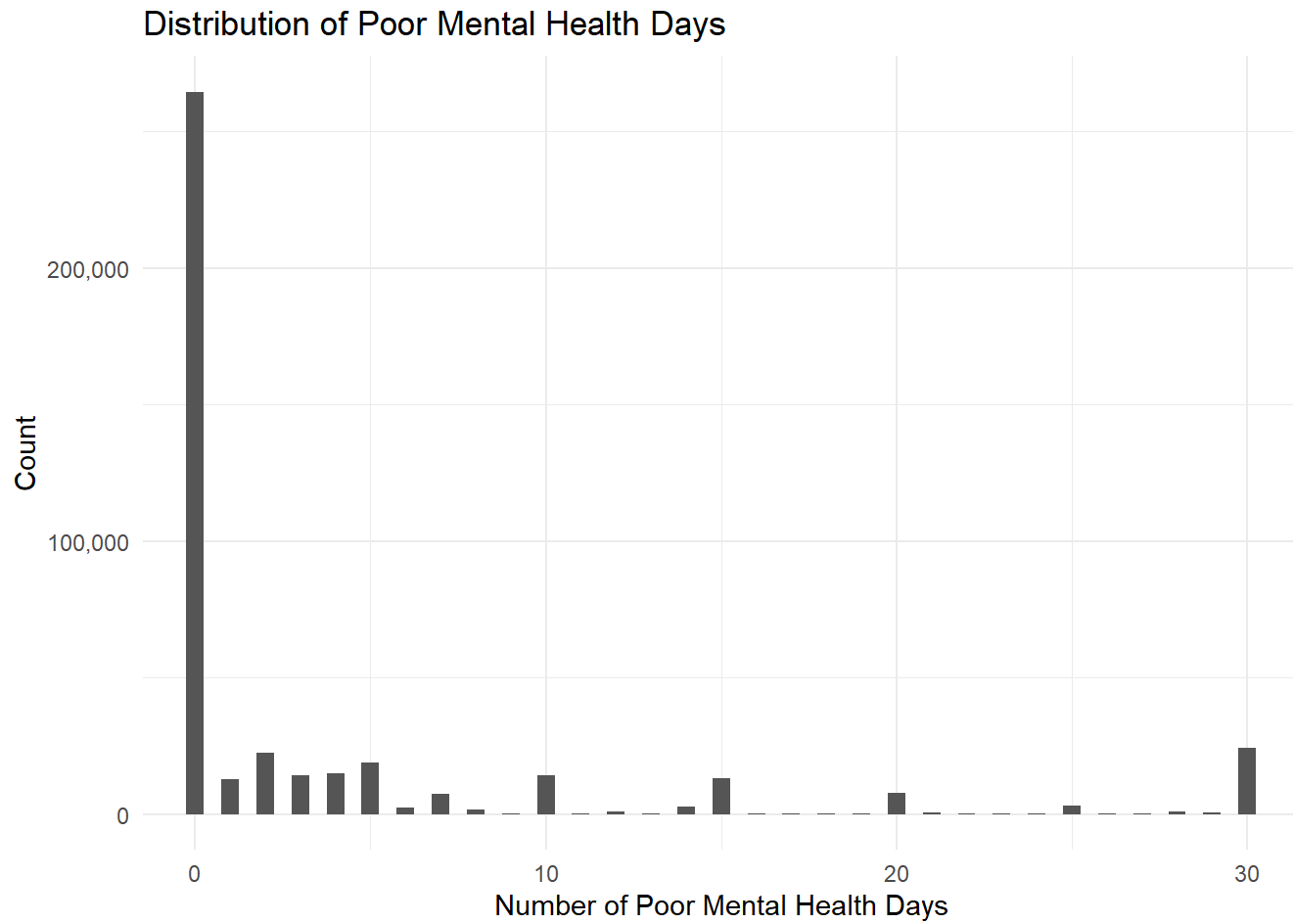
overall health and well-being. Understanding factors that contribute to poor mental health days can enhance prevention and intervention efforts. On a personal note, I am a strong advocate for mental health awareness, which also motivated my decision to focus on factors associated with poor mental health days.

Converting Values -Converted coded categorical variables, EMPLOY1 and EDUCA to factors with new labels. I will also rename the columns in the process, to make them easier to understand. This was done because I will be running regression models and making predictions. This makes the data readable and interpretable for analysis. -Recoded level "88" in quantative variables to value of 0, because it represents a numercal value of 0 days. -Renamed values in the variables to make them shorter, making them quicker to read and analyze. -Created another dataframe after all the cleaning(brf_clean) because this is a shorter name and easier to write syntax. It also indicates that the data is clean and ready for analysis and interpretation.

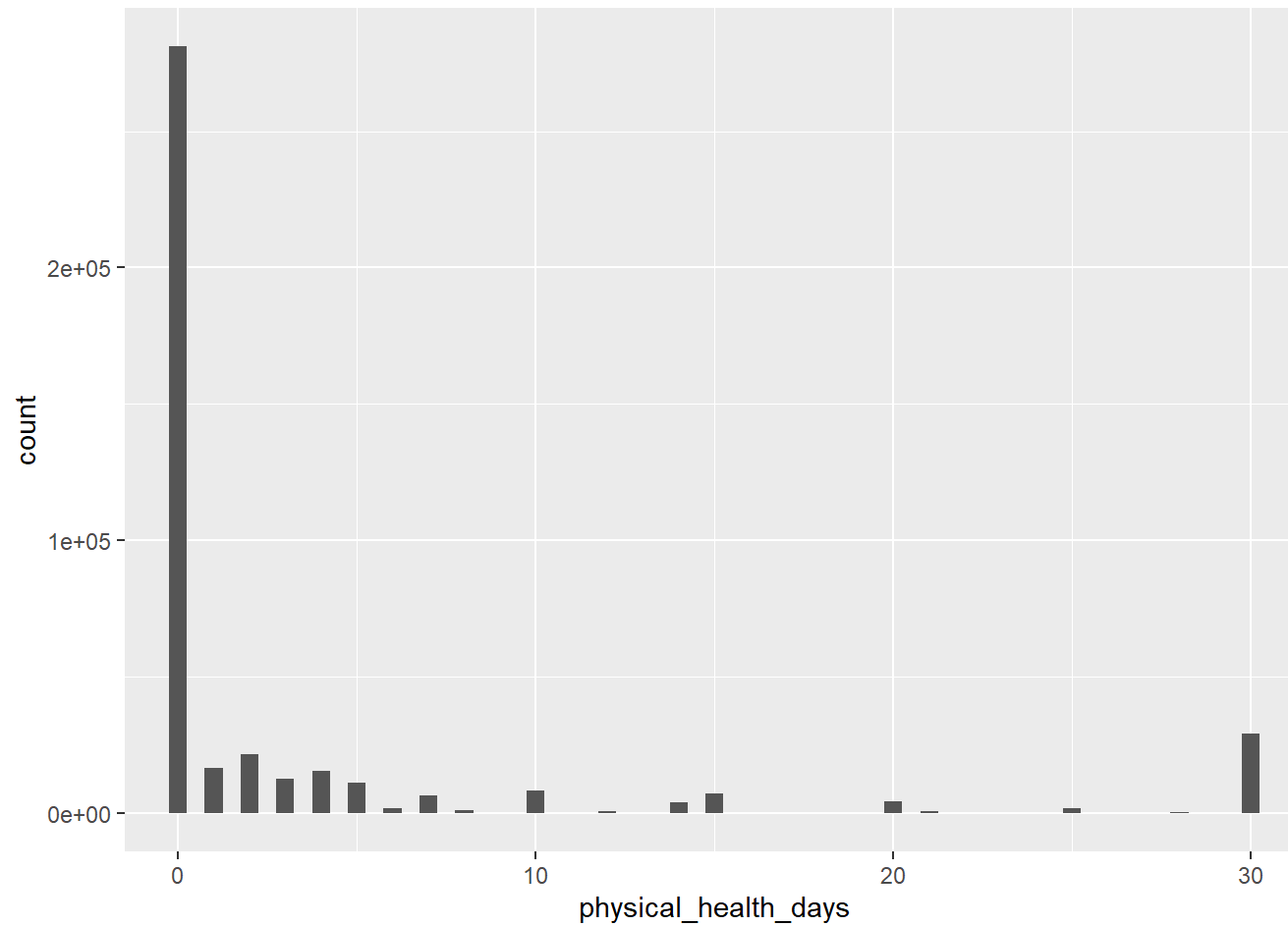
Addressing NAs and missing survey responses -Dropping the NAs and missing survey responses loses 5.5% of the observations. This loss percentage is on the cusp where it could be a problem, so I compared the impact on the data if NAs are removed vs. if they are imputed, to see if either course of action makes a huge impact. -The mean and standard deviation for the MENTHLTH and PHYSHLTH show little change whether the NAs are removed or whether they are imputed. I will work with the dataframe with the imputed values, to preserve the observation size, reduce bias, and increase statistical power. This way I know no values are removed.

```
#mental_health_days Histogram

ggplot(brf_clean, aes(mental_health_days)) +
  geom_histogram(binwidth = .5) +
  coord_cartesian(xlim = c(0, 30)) +
  labs(title = "Distribution of Poor Mental Health Days",
       x = "Number of Poor Mental Health Days",
       y = "Count") +
  scale_y_continuous(labels = scales::comma) +
  theme_minimal()
```



```
#physical_health_days Histogram  
  
ggplot(brf_clean, aes(physical_health_days)) +  
  geom_histogram(binwidth = .5) +  
  coord_cartesian(xlim = c(0, 30))
```

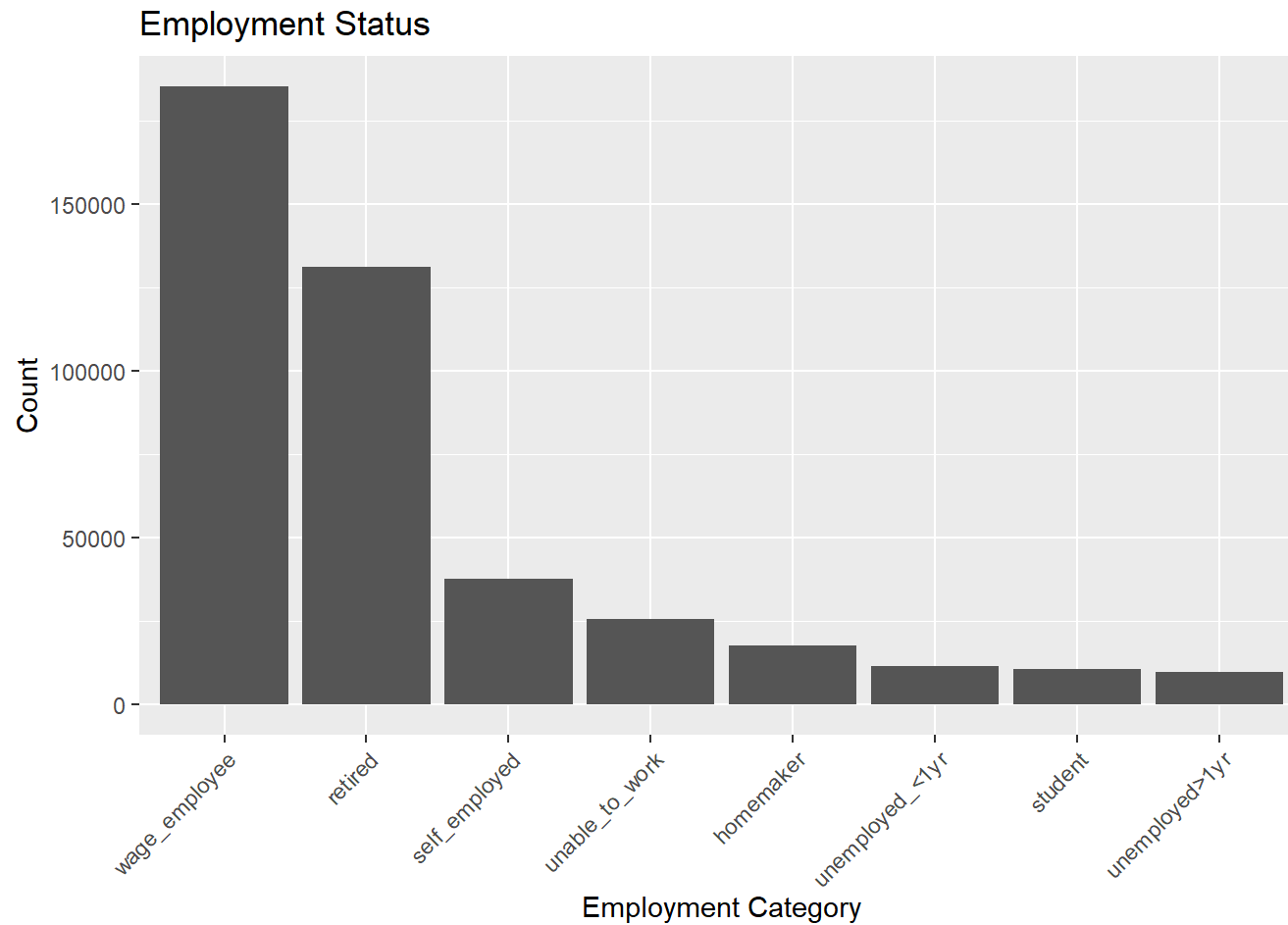


```
labs(title = "Distribution of Poor Physical Health Days",  
      x = "Number of Poor Physical Health Days",  
      y = "Count") +  
scale_y_continuous(labels = scales::comma) +  
theme_minimal()
```

NULL

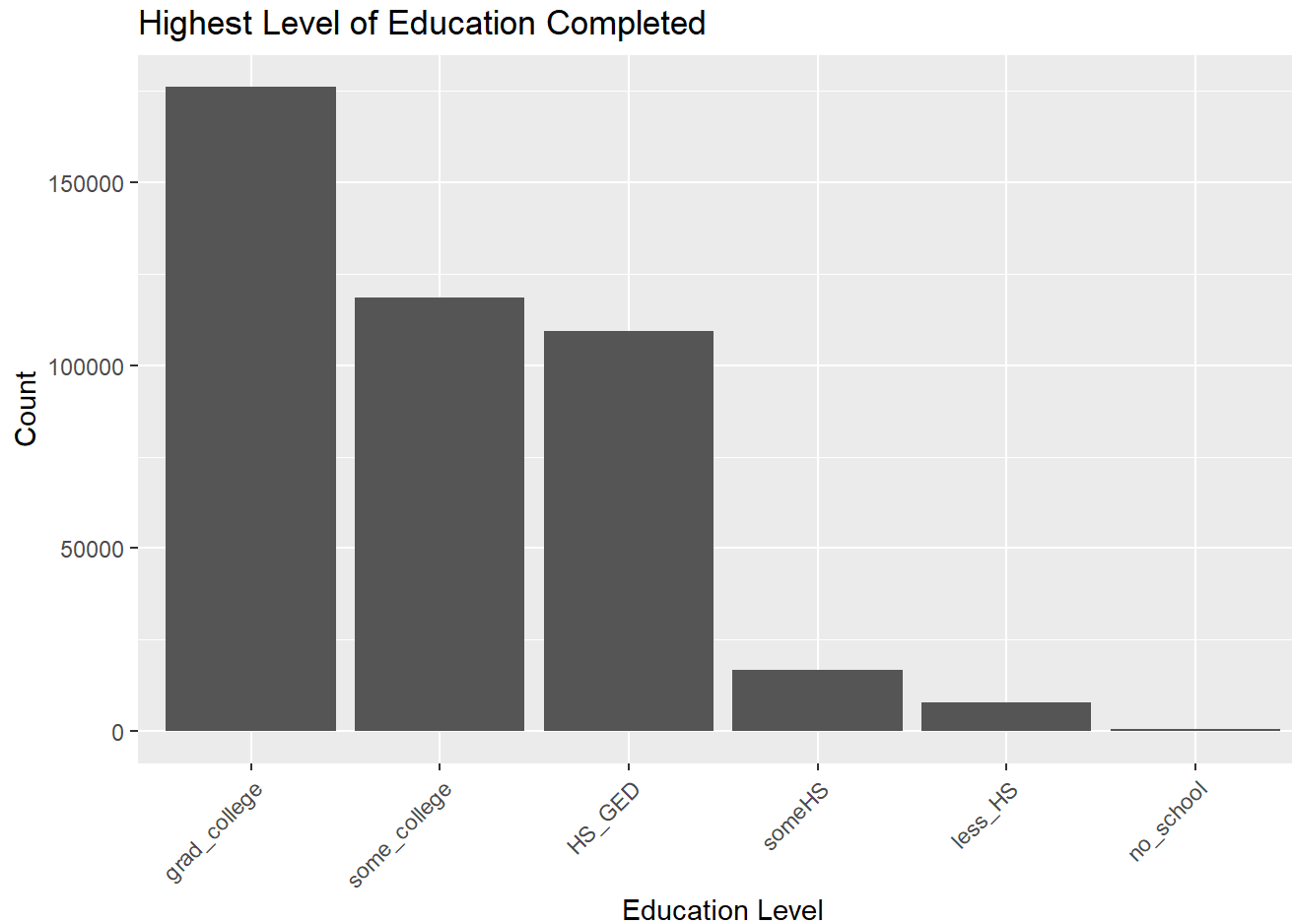
```
#employment status ordered bar chart  
ggplot(brf_clean, aes(x = fct_infreq(employment))) +
```

```
geom_bar() +
labs(title = "Employment Status",
      x = "Employment Category",
      y = "Count") +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
#education level ordered bar chart
ggplot(brf_clean, aes(x = fct_infreq(education))) +
  geom_bar() +
  labs(title = "Highest Level of Education Completed",
        x = "Education Level",
```

```
y = "Count") +  
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

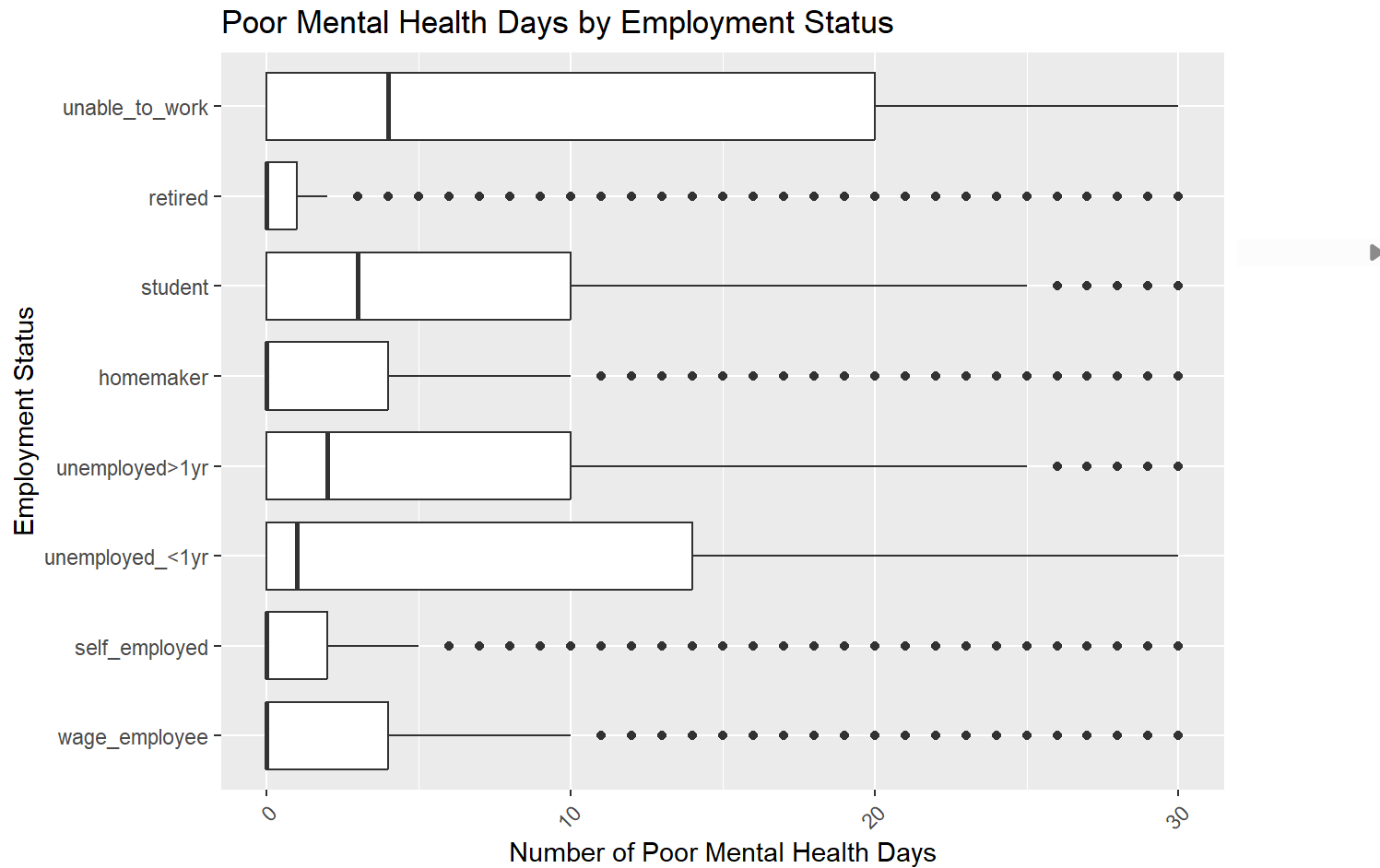


```
#mental_health_days and employment boxplot to identify potential outliers, since I will be using these var  
ggplot(brf_clean, aes(x = employment,  
                      y = mental_health_days)) +  
  geom_boxplot() +  
  coord_flip() +  
  labs(title = "Poor Mental Health Days by Employment Status",  
        x = "Employment Status",
```

```

y = "Number of Poor Mental Health Days") +
theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

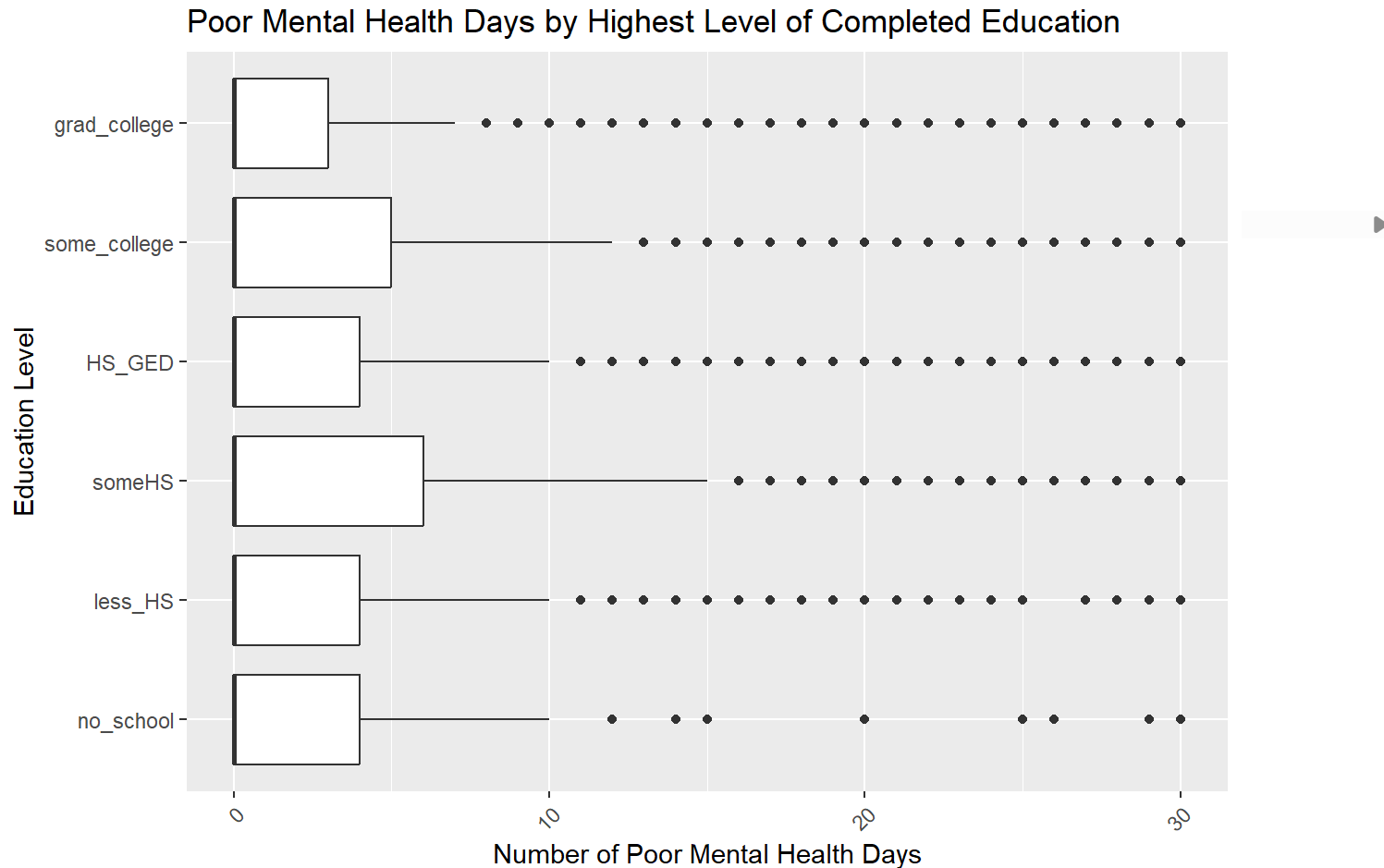


```

#mental_health_days and education boxplot to identify potential outliers, since I will be using these vari
ggplot(brf_clean, aes(x = education,
                      y = mental_health_days)) +
  geom_boxplot() +
  coord_flip() +
  labs(title = "Poor Mental Health Days by Highest Level of Completed Education",
       x = "Education Level",

```

```
y = "Number of Poor Mental Health Days") +  
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Mental Health Days Histogram -This plot shows a right-skewed distribution and are not normally distributed. This shows a large portion of respondents reporting 0 poor mental health days. This also supports that the values of 30 are not errors(harmful outliers), but real reported data. Most respondents report 10 or less poor mental health days per month. *Physical Health Days Histogram* -This plot is very similar to the mental health days plot. This plot shows a right-skewed distribution and are not normally distributed. This shows a large portion of respondents reporting 0 poor mental health days. This also supports that the values of 30 are not errors(harmful outliers), but real reported data. Most respondents report 10 or less poor mental health days per month. *Employment Status Barchart* -This plot

show the counts of the survey responses pertaining to employment status. Wage employees account for the highest of number responses, with retired respondents second highest. The lowest number of responses are people who have been unemployed for over a year. People who have been unemployed under a year, students, and people who have been unemployed for over a year have similar counts. *Highest Level of Education Completed Barchart* -This plot show the counts of the survey responses pertaining to education levels. College graduates account for the highest of number responses, with those who completed some college as the second highest. The lowest number of responses are people who have little to no education. People who have completed some college similar and graduated highschool/GED have similar counts. *Poor Mental Health Days by Employment boxplot* -This plot shows the distribution of poor mental health days across employment status. Those unemployed for less than 1 year and those unable to work show no outliers. Most of their poor mental health days fall in the 3rd quartile. Those who are wage_employees, self_employed, homemakers, students, also show most of their poor mental health days fall in the 3rd quartile and the max value of the outliers is also 30 poor mental health days per month. *Poor Mental Health Days by Highest Level of Completed Education Boxplot* -This plot show the distribution of poor mental health days across education levels. Each education value has outliers where the max value is 30 poor mental health days per month. All values have outliers that are at least 10 or greater poor mental health days per month. Those who completed some high school have the greatest number of responses in the 3rd quartile. Those with little to no school and completed grades 1-8 have the lowest number of responses in the 3d quartile.

```
#Summary Statistics for Quantitative Variables mental_health_days and physical_health_days

#Since the median is near 0, the means i larger than the median, the data for these variables right skewed
var_stats <- describe(brf_clean[, c("mental_health_days", "physical_health_days")]) |>
  as.data.frame()

print(var_stats)
```

	vars	n	mean	sd	median	trimmed	mad	min	max
mental_health_days	1	428767	4.109913	8.093363	0	1.918356	0	0	30
physical_health_days	2	428767	3.808551	8.272191	0	1.420046	0	0	30
	range	skew	kurtosis	se					
mental_health_days	30	2.255694	4.023159	0.0123600					
physical_health_days	30	2.418141	4.560800	0.0126331					


```
#Meaning that 75% of respondents reported 4 or fewer poor mental health days
#75% of respondents reported 3 or fewer poor physical health days
IQR(brf_clean$mental_health_days) #4
```

```
[1] 4
```

```
IQR(brf_clean$physical_health_days) #3
```

```
[1] 3
```

```
#Variance. high values meaning that the data points are spread out over a wide range from the median value
var(brf_clean$mental_health_days) #65.50
```

```
[1] 65.50252
```

```
var(brf_clean$physical_health_days) #68.43
```

```
[1] 68.42915
```

```
#Categorical variables - shows the raw count and percentage of the respondents that falls into each of the
table(brf_clean$employment)
```

wage_employee	self_employed	unemployed<1yr	unemployed>1yr	homemaker
185333	37620	11356	9686	17501
student	retired	unable_to_work		
10619	131088	25564		

```
round(prop.table(table(brf_clean$employment)) * 100, 1)
```

wage_employee	self_employed	unemployed<1yr	unemployed>1yr	homemaker
43.2	8.8	2.6	2.3	4.1

student	retired	unable_to_work
2.5	30.6	6.0

```
#education rounded to 2 decimal places, converted to a percentage
table(brf_clean$education)
```

no_school	less_HS	someHS	HS_GED	some_college	grad_college
580	8017	16783	109259	118290	175838

```
round(prop.table(table(brf_clean$education)) *100, 1)
```

no_school	less_HS	someHS	HS_GED	some_college	grad_college
0.1	1.9	3.9	25.5	27.6	41.0

```
#mental_health_days and employment status
mental_health_descriptive_stat_1 <- brf_clean |>
  group_by(employment) |>
  summarise(
    mean_mental_health_days = round(mean(mental_health_days, na.rm = TRUE), 2),
    sd_mental_health_days = round(sd(mental_health_days, na.rm = TRUE), 2),
    n = n()
  )

print(mental_health_descriptive_stat_1)
```

A tibble: 8 × 4

employment	mean_mental_health_days	sd_mental_health_days	n
<fct>	<dbl>	<dbl>	<int>
1 wage_employee	3.98	7.6	185333
2 self_employed	3.2	7.2	37620
3 unemployed<1yr	7.54	10.7	11356
4 unemployed>1yr	7.19	10.2	9686
5 homemaker	4.2	8.11	17501
6 student	6.93	8.84	10619

7 retired	2.59	6.69	131088
8 unable_to_work	10.3	11.9	25564

I decided to keep the outliers in the dataset because the survey already has a built-in cap at 30 days(which is numerical), so the values can't go beyond that limit. The responses above the 3rd quartile (which are the seeming outliers) cutoff (around 22.5 days of poor mental or physical health) aren't errors or anomalies. They reflect people who genuinely reported experiencing a high number of poor health days.

Removing those responses would take out valid data and could bias the results by underrepresenting people who are struggling more. Since these values accurately represent real survey respondents, they will remain in to avoid skewing the findings.

Summary Statistics Discussion for Quantitative Variables 75% of respondents reported a mean of 4 or fewer poor mental health days 75% of respondents reported a mean of 3 or fewer poor physical health days

Variance Values are high for these variables. This makes sense because the plots show that the data points are spread out over a wide range from the median value. poor mental health days = 65.50 physical health days = 68.43

Categorical Variables Summary Statistics Employment: About 43% of respondents are wage earners. 30% retired About 9% self-employed 6% unable to work About 4% homemakers About 3% are unemployed less than 1 yr About 2% are unemployed over a year About 3% are students Education: 41% graduated college or more About 28% some college About 26% graduated high school or have a GED About 4% have some high school education About 2% less than high school Less than 1% have little to no education

Mean of Poor Mental Health Days by Employment Status(rounded to nearest whole number): Wage earner = 4 days Self employed = 3 days Unemployed less than 1 year = 8 days Unemployed more than 1 year = 7 days Homemaker = 4 days Student = 7 days Retired = 3 days Unable to work = 10 days

The standard deviation of these variables ranges from approximately 6 - 11. This indicates that the spread of data points around the mean varies across the variables, which aligns with that boxplot visualizations.

```
#mental_health_days ~ employment
#I did not standarize because my I'm only using 1 predictor variable and it is categorical
model1 <- lm(mental_health_days ~ education + physical_health_days, data = brf_clean)
```

```
print(model1)
```

Call:

```
lm(formula = mental_health_days ~ education + physical_health_days,  
    data = brf_clean)
```

Coefficients:

(Intercept)	educationless_HS	educationsomeHS
2.7260	-0.2610	1.1785
educationHS_GED	educationsome_college	educationgrad_college
0.4124	0.6967	-0.1486
physical_health_days		
0.2905		

```
#Regression model2  
model2 <- lm(mental_health_days ~ employment + education,  
             data = brf_clean)  
  
print(model2)
```

Call:

```
lm(formula = mental_health_days ~ employment + education, data = brf_clean)
```

Coefficients:

(Intercept)	employmentself_employed
3.1590	-0.8137
employmentunemployed_<1yr	employmentunemployed>1yr
3.4062	3.0673
employmentthomemaker	employmentstudent
0.1168	2.7447
employmentretired	employmentunable_to_work
-1.4531	6.0886
educationless_HS	educationsomeHS
0.1516	1.4542

educationHS_GED	educationsome_college
1.0041	1.3656
educationgrad_college	
0.4364	

```
#Employment and education are statistically significant predictors of poor mental health days; however, th

#Regression model3

model3 <- lm(mental_health_days ~ employment + physical_health_days, data = brf_clean)

print(model3)
```

Call:

```
lm(formula = mental_health_days ~ employment + physical_health_days,
    data = brf_clean)
```

Coefficients:

(Intercept)	employmentself_employed
3.3981	-0.8621
employmentunemployed_<1yr	employmentunemployed>1yr
2.5509	2.5898
employmenthomemaker	employmentstudent
-0.1819	2.9427
employmentretired	employmentunable_to_work
-2.0359	3.0533
physical_health_days	
0.2721	

```
#Summary of all 3 models for analysis
summary(model1)
```

Call:

```
lm(formula = mental_health_days ~ education + physical_health_days,
```

```
data = brf_clean)
```

Residuals:

Min	1Q	Median	3Q	Max
-12.6188	-3.4228	-2.5774	0.2607	27.5349

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.726030	0.320001	8.519	< 2e-16 ***
educationless_HS	-0.260957	0.331306	-0.788	0.430896
educationsomeHS	1.178523	0.325419	3.622	0.000293 ***
educationHS_GED	0.412369	0.320777	1.286	0.198607
educationsome_college	0.696721	0.320713	2.172	0.029825 *
educationgrad_college	-0.148629	0.320469	-0.464	0.642801
physical_health_days	0.290475	0.001434	202.591	< 2e-16 ***

Signif. codes:	0 '***'	0.001 '**'	0.01 '*'	0.05 '.' 0.1 ' ' 1

Residual standard error: 7.705 on 428760 degrees of freedom

Multiple R-squared: 0.0937, Adjusted R-squared: 0.09369

F-statistic: 7388 on 6 and 428760 DF, p-value: < 2.2e-16

```
summary(model2)
```

Call:

```
lm(formula = mental_health_days ~ employment + education, data = brf_clean)
```

Residuals:

Min	1Q	Median	3Q	Max
-10.7017	-3.5953	-2.7100	-0.1422	28.2941

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.15898	0.32622	9.684	< 2e-16 ***
employmentself_employed	-0.81369	0.04439	-18.329	< 2e-16 ***
employmentunemployed_<1yr	3.40618	0.07615	44.729	< 2e-16 ***

employmentunemployed>1yr	3.06735	0.08198	37.417	< 2e-16	***
employmenthomemaker	0.11679	0.06249	1.869	0.06162	.
employmentstudent	2.74470	0.07851	34.960	< 2e-16	***
employmentretired	-1.45313	0.02840	-51.174	< 2e-16	***
employmentunable_to_work	6.08858	0.05348	113.838	< 2e-16	***
educationless_HS	0.15160	0.33735	0.449	0.65316	
educationsomeHS	1.45417	0.33136	4.388	1.14e-05	***
educationHS_GED	1.00411	0.32669	3.074	0.00211	**
educationsome_college	1.36562	0.32665	4.181	2.91e-05	***
educationgrad_college	0.43636	0.32645	1.337	0.18132	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.845 on 428754 degrees of freedom

Multiple R-squared: 0.06039, Adjusted R-squared: 0.06036

F-statistic: 2296 on 12 and 428754 DF, p-value: < 2.2e-16

```
summary(model3)
```

Call:

```
lm(formula = mental_health_days ~ employment + physical_health_days,
    data = brf_clean)
```

Residuals:

Min	1Q	Median	3Q	Max
-14.6145	-3.3981	-1.9879	0.0936	28.6378

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.398124	0.017860	190.269	< 2e-16 ***
employmentself_employed	-0.862125	0.042787	-20.149	< 2e-16 ***
employmentunemployed<1yr	2.550898	0.073349	34.777	< 2e-16 ***
employmentunemployed>1yr	2.589796	0.078931	32.811	< 2e-16 ***
employmenthomemaker	-0.181943	0.059872	-3.039	0.00237 **
employmentstudent	2.942712	0.075495	38.979	< 2e-16 ***
employmentretired	-2.035929	0.027529	-73.956	< 2e-16 ***

```
employmentunable_to_work    3.053300    0.053515  57.055 < 2e-16 ***
physical_health_days          0.272103    0.001486 183.055 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 7.566 on 428758 degrees of freedom
Multiple R-squared:  0.1261,    Adjusted R-squared:  0.1261
F-statistic: 7733 on 8 and 428758 DF,  p-value: < 2.2e-16
```

```
#Assessing best model fit
```

```
#Adjusted R-squared because if I use r-squared with models with multiple predictors, like I have, it can c
```

```
summary(model1)$adj.r.squared # 0.09
```

```
[1] 0.09369007
```

```
summary(model2)$adj.r.squared # 0.06
```

```
[1] 0.06036343
```

```
summary(model3)$adj.r.squared # 0.12 best model so far (highest adjusted r-squared value). This makes sens
```

```
[1] 0.1260835
```

```
#AIC to determine the best model
```

```
AIC(model1) #2967759
```

```
[1] 2967759
```

```
AIC(model2) #2983249
```

```
[1] 2983249
```



```
AIC(model3) #2952156 Again, model3 is the best fit because it has the lowest AIC
```

```
[1] 2952156
```

```
#BIC to determine the best model fit, because this is useful when you have a larger sample size, which we
BIC(model1) #2967847
```

```
[1] 2967847
```

```
BIC(model2) #2983402
```

```
[1] 2983402
```

```
BIC(model3) #2952265 Here model3 is again showing the best fit model.
```

```
[1] 2952265
```

```
#Check for multicollinearity using VIF because I am using 3 or more predictor variables. If I only had two
vif(model1)
```

	GVIF	Df	GVIF^(1/(2*Df))
education	1.016031	5	1.001592
physical_health_days	1.016031	1	1.007984

```
vif(model2)
```

	GVIF	Df	GVIF^(1/(2*Df))
employment	1.067706	7	1.004690
education	1.067706	5	1.006573

```
vif(model3)
```

	GVIF	Df	GVIF ^{1/(2*Df)}
employment	1.1325	7	1.008927
physical_health_days	1.1325	1	1.064190

Q14 Discussion:

Interpretation of Model1 (education + physical_health_days are predictors) - The intercept of 3 (rounded to nearest whole number) is the predicted number of poor mental health days of a person who completed little to no education(the reference group), when poor physical health days = 0. For each additional poor physical health day that that is reported, the poor mental health days will increased by 0.29 days on average, holding education constant.

-The education variable values "less_HS", "HS_GED", and "grad_college" are not significantly significant, as illustrated by the significance codes. This means that these values do not provide enough evidence to conclude that they have an effect on the number of reported poor mental health days.

- Since the intercept represents the baseline prediction, it corresponds to the expected number of poor mental health days for an individual in the lowest education group (the reference category). The results show that a person with:
 1. some high school = will report 1 poor mental health day more than someone with little to no education.
 2. some college = will report about 0.70 more poor mental health days than someone with little to no education. -
 Overall, education level doesn't seem to make as much of an impact on someone's mental health.

Interpretation of Model2 (education + employment are predictors) - The intercept in this model, rounded to nearest whole-number,ndicates that wage employees with little to no education (the reference category) report roughly 3 poor mental health days per month. - Homemakers, college graduates, and those who only completed grades 1-8, The values homemaker, less_HS, and grad_college are not statistically significant meaning there is insufficient evidence that they affect the number of reported poor mental health days. -Employment effects 1. Self-employed individuals report fewer than 1 poor mental health days than wage_employees. 2. Recently unemployed individuals report about 3 more poor mental health days than wage_employees 3. Long-term unemployed individuals report about 3 more poor mental health days than wage_employees. 4. Students report about 3 more days more poor mental health days than wage_employees. 5. Retired individuals report about 1.45 fewer poor mental health days than wage_employees. 6. Individuals unable to work report about 6 more poor mental health days than

wage_employees. -Education Effects 1. Respondents with some highschool, some college and highschool graduates, report around 1 more poor mental health day than someone with little to no education.

Employment status has more of an impact on poor mental health days than education level.

****Interpretation of Model3 (employment + physical_health_days are predictors)*** -The intercept is the wage_employee value which means those who fall into that employment category report approximately 3 days of poor mental health per month. -For each additional poor physical health day, mental health days increase by about 0.27 days. -All the employment values are significantly significant. -Employment effects 1. Self-employed individuals report less than 1 fewer poor mental health days than wage_employees. 2. Individuals unemployed for less than one year report about 2.5 more poor mental health days than wage_employees. 3. Individuals unemployed for more than one year report 2.5 more poor mental health days than wage_employees. 4. Homemakers report less than 1 poor mental health day than wage_employees. 5. Students report about 3 more poor mental health days than wage_employees. 6. Retired individuals report about 2 fewer poor mental health days than wage_employees. 7. Individuals unable to work report about 3 more poor mental health days than wage_employees. This is about the same for students. -Overall, unemployed and individuals who are unable to work are associated with worse mental health. Retirement is associated with better mental health, which makes sense! Physical health remains a major factor of mental health state. -Physical health

Model Diagnostics and Interpretation Adjusted R-squared Values and interpretation I will be looking for the highest value because adjusted r-squared penalizes a model if it has unhelpful predictors, so having a higher value indicates less penalization, meaning the model with the most helpful predictors. Therefore model3 is the best performing because it has the highest r-squared value. Model1 = 0.09 Model2 = 0.06 Model3 = 0.12

#AIC Values and Interpretation Here I am looking for the lowest value, because a low value indicates balanced model accuracy with parsimony, which minimizes information loss. Therefore model 3 is, again, the best model because it has the lowest AIC Model1 = 2967759 Model2 = 2983249 Model3 = 2952156

BIC Values and Interpretation Want the lowest value because that indicates the model is the most parsimonious(simplest). Model3 has the lowest value and is the best model. Model1 = 2967847 Model2 = 2983402 Model3 = 2952265

VIF Values and Interpretation -Check for multicollinearity using VIF Values because I am using 3 or more predictor variables. If I only had two predictor variables, I would use correlation matrix. After completing outside research, I learned that I interpret the $GVIF^{(1/(2Df))}$ values because I am using categorical variables with multiple degrees of

freedom, I only interpret the adjusted $GVIF^{(1/(2Df))}$ value, which standardizes GVIF, so I can compare across the predictors. I still use the same VIF cut-off rule, that any $GVIF^{(1/(2*Df))}$ value > 5 indicates there is no multicollinearity. All the values in this matrix are around 1(rounded to 2 decimals places), therefore there is no multicollinearity between education, physical_health_days, and education.

Model1 = education 1.00, physical_health_days 1.01 Model2 = employment 1.07, education 1.01 Model3 = employment 1.01, 1.06

```
#Model3 is the best fit
new_individuals <- data.frame(
  employment = c("wage_employee", "self_employed", "retired"),
  education = c("less_HS", "HS_GED", "grad_college"),
  physical_health_days = c(0, 5, 10)
)
#change to factors
new_individuals$employment <- factor(
  new_individuals$employment,
  levels = levels(brf_clean$employment)
)

new_individuals$education <- factor(
  new_individuals$education,
  levels = levels(brf_clean$education)
)

predictions<- predict(model3, newdata = new_individuals)
#renamed columns for faster interpretation and rounded to to the nearest whole number because mental health
names(predictions) <- c("Person1", "Person2", "Person3")

predictions <- round(predictions, 0)

print(new_individuals)
```

	employment	education	physical_health_days
1	wage_employee	less_HS	0

2	self_employed	HS_GED	5
3	retired	grad_college	10

```
print(predictions)
```

Person1	Person2	Person3
3	4	4

Discussion:

These predictions can be interpreted as the following:

Person1: A wage employee, with a less than a high school education, and reports 0 poor physical health days, is predicted to experience approximately 3 poor mental health days per month.

Person2: A self-employed individual, who completed high school/GED, and reported 5 poor physical health days is predicted to experience approximately 4 poor mental health days per month.

Person3: A retired individual, who is a college graduate, and reported 10 poor physical health is predicted to experience approximately 4 poor mental health days.

Although the individuals differ in employment status, education level, and number of poor physical health days, their predicted poor mental health days are relatively similar (around 3–4 days per month). This suggests that these factors, at least in this model, do not lead to large differences in predicted mental health outcomes. However, a person who is retired, regardless of their physical health, is predicted to have fewer poor mental health days (which also accounts for those outliers). (Goes to show that the mind is stronger than the body!)

As with any self-reported data (survey) reliability of this data depends on how truthfully or how accurately respondents reported their poor mental and physical health days, education level, and employment status.

Limitations of the model include the absence of other factors that could impact a person's reported poor mental health days such as income, age, gender, race, marital status, etc. These predictors only account for a small portion of the variation in mental health days.
